

How to automate result reporting

September 14, 2026

<https://github.com/jimb0w/automation>

Correspondence to:

Jedidiah Morton

Jedidiah.Morton@monash.edu

Research Fellow

Baker Heart and Diabetes Institute, Melbourne, Australia

Monash University, Melbourne, Australia

Contents

1	Data cleaning	2
---	---------------	---

1 Data cleaning

Document everything.

Say what you're going to do, do it, then comment on it.

I'm going to create a random health dataset with the following variables:

- Sex (0 = female; 1 = male)
- Age (in years)
- Diabetes status (0 = no diabetes; 1 = diabetes)
- Smoking status (0 = non-smoker; 1 = smoker)
- LDL-C (in mmol/L)
- Outcome (in outcome units)

As so:

```
cd /home/jimb0w/Downloads/tempauto/  
clear  
set obs 20000  
set seed 1312  
gen sex = runiform(0,1)  
gen age = rnormal(60,15)  
gen diabetes = runiformint(0,1)  
gen smoking = runiformint(0,1)  
gen ldl = rnormal(3,0.5)  
gen outcome = rnormal(200,30)+diabetes*rnormal(100,20)  
export delimited using randomhealthdata.csv, replace
```

Now I'm going to check the variables make sense:

```
. su(sex), detail
```

sex				
	Percentiles	Smallest		
1%	.0096901	.0000432		
5%	.0501992	.0001577		
10%	.1017735	.0001719	Obs	20,000
25%	.2508185	.0001787	Sum of wgt.	20,000
50%	.4990473		Mean	.4991676
		Largest	Std. dev.	.2875379
75%	.7493168	.9998247		
90%	.8966301	.9998952	Variance	.0826781
95%	.947588	.9999172	Skewness	.0015478
99%	.9893163	.9999394	Kurtosis	1.805996

```
. su(diab), detail
```

diabetes				
	Percentiles	Smallest		
1%	0	0		
5%	0	0		
10%	0	0	Obs	20,000
25%	0	0	Sum of wgt.	20,000
50%	0		Mean	.494
		Largest	Std. dev.	.4999765
75%	1	1		
90%	1	1	Variance	.2499765
95%	1	1	Skewness	.0240017

```

99%          1          1      Kurtosis      1.000576
. su(smoking), detail

      smoking
-----
Percentiles      Smallest
 1%             0             0
 5%             0             0
10%             0             0      Obs             20,000
25%             0             0      Sum of wgt.        20,000
50%             1
75%             1      Largest
90%             1             1      Std. dev.         .5000118
95%             1             1      Variance          .2500118
99%             1             1      Skewness          -.0034
                                Kurtosis          1.000012
. su(age), detail

      age
-----
Percentiles      Smallest
 1%      24.66596      .835979
 5%      35.15182      1.354185
10%      40.73132      4.960024      Obs             20,000
25%      49.76038      5.434331      Sum of wgt.        20,000
50%      59.77124
75%      69.96964      Largest
90%      78.95889      109.9906      Std. dev.         59.82388
95%      84.55445      111.0537      Variance          14.99574
99%      94.6009      111.059      Skewness          224.8723
                                Kurtosis          - .0096071
                                Kurtosis          3.032043
. su(ldl), detail

      ldl
-----
Percentiles      Smallest
 1%       1.81618      1.169115
 5%       2.166875      1.17411
10%       2.354322      1.184871      Obs             20,000
25%       2.66082      1.20128      Sum of wgt.        20,000
50%       3.000187
75%       3.334033      Largest
90%       3.636441      4.78299      Std. dev.         2.997539
95%       3.817179      4.849288      Variance          .50188
99%       4.168352      4.876576      Skewness          .2518836
                                Kurtosis          -.0133279
                                Kurtosis          3.031805
. su(outcome), detail

      outcome
-----
Percentiles      Smallest
 1%      139.2732      87.21138
 5%      162.0865      91.83277
10%      174.7943      91.93818      Obs             20,000
25%      199.8294      92.32668      Sum of wgt.        20,000
50%      244.0778
75%      299.1514      Largest
90%      329.6298      426.0753      Std. dev.         249.3994
95%      345.9959      430.0988      Variance          59.80981
99%      374.6187      434.5064      Skewness          3577.214
                                Kurtosis          .163877
                                Kurtosis          2.073516

```

It looks like these variables make sense and there don't appear to be any errors with the data. But we should check further: let's produce a histogram to check the continuous variables. The code below produces Figure 1.1.

```
mkdir GPH
```

```

hist ldl, frequency ///
xtitle("LDL-C (mmol/L)")
graph save GPH/LDL_hist, replace
hist age, frequency ///
xtitle("Age (years)")
graph save GPH/Age_hist, replace
hist outcome, frequency ///
xtitle("Outcome (units)")
graph save GPH/Outcome_hist, replace
graph combine ///
GPH/LDL_hist.gph ///
GPH/Age_hist.gph ///
GPH/Outcome_hist.gph ///
, cols(1) altshrink xsize(3)
graph export GPH/distcheck.pdf, as(pdf) replace

```

Figure 1.1: Histograms of LDL-C, Age, and outcome.

